Help for command ReportGCP

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1 Descrition of ReportGCP

This command generate report on quality of orientation relative to a set of Ground control point. It generate two kind of information :

- statisical information in `csv` format , these file are generated in the ReportDir of the application see section 2;
- visual information, in `jpg/tiff` format, that contains a visualization of the residual for each camera an, optionnaly, for each image, these result are stored in the VisuDir of the application, see section 3;

2 Generated report as `csv` files by ReportGCP

The following csv files are generated :

- `ByCam.csv` : residual of projection in pixel, agregated by camera, usable to analyse internal calibration quality
- `ByGCP.csv` : residual of projection in pixel, agregated by 3D point, usable to analyse ground coordinates quality
- `ByGCP3D.csv` : difference in ground coordinate between 3D coordinate and budle intesection
- `ByGCP3D_Stat.csv` : statistique on value of `ByGCP3D.csv` (average ...)
- `ByImage.csv` : residual of projection in pixel, agregated by images, usable to analyse pose estimation quality
- `Detail.csv` : residual of projection in pixel for each point/image, usable to analyse outliers

3 Generated visualisation files by ReportGCP

For each camera *Cam*, the following low resolution images are generated to anlyse the distribution of GCP on the sensor and the bias of reprojection :

- `W_RawResidual_`*Cam*: raw distribution of point on the sensor plane (generally sparse)

- `W_FiltResidual_`Cam : gaussian filtered distribution of point on the sensor plane to make it continuous
- `X_Residual_`Cam and `Y_Residual_`Cam images of reprojection error, gaussian filtered, usable to detect spatial bias in intrinsic parameters.

When the optional vector `GFV` [1] is used, a visualisation is generated for each image ; it shows for each GCP if it is detected and, if it is, shows the vector between detection and projection .

- let note `GFV`$= [M, W, \rho, Z? = 2, Jpg? = 1]$ the vector of size 3 to 5;
- for each *detected* GCP, a vector between its localisation and its projection is drawn with an amplification of M and a witdh W
- for each *undected* GGP, a circle of ray ρ is drawn centereed on projection of the GCP
- Z is the reduction factor and Jpg is used a boolean indicating if `Jpeg/Tiff` is to be used.

[1] generate field vecror